

nology. Vibrant stained glass windows in European cathedrals owed their rich colours to nanoparticles of gold chloride and other metal oxides and chlorides: Ruby Red is due to gold nanoparticles while Deep Yellow is due to silver nanoparticles. It is the size of these particles that produced the colour variations and as our modern scientists will agree, the dramatic change in material properties at the nano-scale is a vital component of nanotechnology. Research has also shown that nanoparticles of gold, when activated by sunlight, can destroy air-borne pollutants such as methanol and other volatile organic chemicals



Stained Glass Windows in a Church

The 9th century AD Abbasid potters in what is today Iraq, used a decorative technique called lusterware, which involves using a lead-based glaze to create a golden shine on a pot without gold in it. Although the potters considered it to be true alchemy, we know that they owed it to the silver or copper or other metallic nanoparticles.



Abbasid Pottery

Renaissance Period (1450AD – 1600AD)

Artisans colouring pottery in Derula, Umbria were also practicing a similar form of nanotechnology, like their Abbasian counterparts. Derula ceramicists were producing iridescent and metallic glazes by using nanoparticles of copper and silver metal. Instead of scattering light, these particles caused light to bounce off their surface at different wavelengths, giving them the high-in-demand metallic effects.



Damascus Saber Blades

The “Damascus” saber blades contained carbon nanotubes and cementite nanowires—an ultrahigh-carbon steel formulation that gave them strength,

resilience, the ability to hold a keen edge, and a visible moiré pattern in the steel that give the blades their name.